Asymptotic-Enriched Physics-Informed Neural Networks for Reentrant Corner Flows

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Abstract

PINN can struggle with corner flows because the network tries to learn simultaneously: the smooth bulk solution, the singular stress field, and the boundary layer structure. This usually causes spectral bias and poor convergence near reentrant corners. In this work we propose an improvement on the PINN methodology including asymptotic expressions for constructing the loss function. According to our tests conducted on a Newtonian fluid flow, the proposed scheme can enhance the accuracy of the PINN solutions around the reentrant corner.

Keywords: physics-informed, singularity, reentrant corner flow, asymptotics.

1. Introduction

2. Governing equations

The governing equations considered in this study are the mass and momentum conservation equations, expressed as follows:

$$\begin{aligned} &\nabla \cdot \mathbf{v} = 0, \\ &\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\,\mathbf{v} \right) = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \end{aligned} \tag{1}$$

where $\mathbf{v}$ is the fluid velocity, p is the pressure, ρ is the fluid density, and $\boldsymbol{\tau}$ is the extra-stress tensor. For Newtonian fluids, the extra-stress tensor is given by

$$\boldsymbol{\tau} = 2\mu\mathbf{D}, \tag{2}$$

where μ is the dynamic viscosity and $\mathbf{D} = \frac{1}{2}\left(\nabla\mathbf{v} + \nabla\mathbf{v}^{\intercal}\right)$ is the rate of strain tensor.

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3. Conclusions

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